

**A Machine Learning Model for Multi-Class Classification of Quenched
and Partitioned Steel Microstructure Type by the k-Nearest Neighbor
Algorithm**

Minor Project Report

submitted for partial fulfilment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY
IN
INDUSTRIAL AND PRODUCTION ENGINEERING

Under the guidance of

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CERTIFICATE

This is to certify that the Minor Project Report titled, 'A Machine Learning Model for Multi-Class Classification of Quenched and Partitioned Steel Microstructure Type by the k-Nearest Neighbor Algorithm' submitted by Aditya Kumar Yadav (23113006), Viraj Singh (23113076), and Vishal Mishra (23113077) to Dr B R Ambedkar National Institute of Technology Jalandhar, Punjab, India, in partial fulfilment of the requirements for the award of the degree in Industrial and Production Engineering, is an authentic record of the project work carried out under the supervision of Dr. Aviral Mishra during the academic year 2025–2026.

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ACKNOWLEDGEMENTS

We express our sincere gratitude to our project supervisor, **Dr. Aviral Mishra**, Department of Industrial and Production Engineering, Dr B R Ambedkar National Institute of Technology Jalandhar, for his invaluable guidance, constant support, and encouragement throughout the duration of this minor project. His expertise in the field of materials science and machine learning has been instrumental in shaping the direction of this study.

We also extend our thanks to the faculty and staff of the Department of Industrial and Production Engineering for providing the necessary academic resources and a conducive learning environment. We are grateful to the authors of the original research paper — Gupta, Chakroborty, Ghosh, and Ganguly — whose work on k-NN-based classification of Q&P steel microstructures formed the foundation of this study.

Finally, we thank our families and friends for their moral support and encouragement throughout this endeavour.

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May 2026

LIST OF SYMBOLS AND ABBREVIATIONS

Symbol / Abbreviation	Description	Unit
Q&P	Quenching and Partitioning	–
k-NN	k-Nearest Neighbor	–
ML	Machine Learning	–
AHSS	Advanced High-Strength Steel	–
M	Martensite	–
RA	Retained Austenite	–
B	Bainite	–
F	Ferrite	–
C	Carbide	–
Ac1	Lower Critical Temperature	°C
Ac3	Upper Critical Temperature	°C
Ms	Martensite Start Temperature	°C
QT	Quenching Temperature	°C
PT	Partitioning Temperature	°C
TMAE	Total Micro-Alloying Elements (Nb+V+Ti)	wt.%
SMOTE	Synthetic Minority Over-sampling Technique	–
SVM	Support Vector Machine	–
SEM	Scanning Electron Microscope	–
TP	True Positive	–
TN	True Negative	–
FP	False Positive	–
FN	False Negative	–
HRC	Hardness (Rockwell C Scale)	–
PAGB	Prior Austenite Grain Boundary	–

ABSTRACT

This project report presents a comprehensive review and study of a machine learning-based approach for the multi-class classification of Quenched and Partitioned (Q&P) steel microstructure types using the k-Nearest Neighbor (k-NN) algorithm. Q&P steel is a new-generation Advanced High-Strength Steel (AHSS) that offers an improved combination of strength and ductility owing to its retained austenite-containing multiphase microstructure. Predicting the type of microstructure that evolves under specific compositional and heat treatment conditions is a critical yet complex task in metallurgical design.

The study is based on a dataset of 348 Q&P steel samples compiled from approximately 107 research articles, characterised by nine input features: carbon content (C wt.%), silicon content (Si wt.%), manganese content (Mn wt.%), total micro-alloying elements (TMAE wt.%), lower critical temperature (Ac1), upper critical temperature (Ac3), martensite start temperature (Ms), quenching temperature (QT), and partitioning temperature (PT). The target variable represents six microstructure types: {M, B, RA}, {M, F, B, RA}, {M, F, RA}, {M, F, RA, C}, {M, RA}, and {M, RA, C}.

Data preprocessing involved feature standardisation to zero mean and application of the Synthetic Minority Over-sampling Technique (SMOTE) to address class imbalance. The k-NN classifier was trained on a 70/30 train-test split and optimised via hyperparameter tuning, yielding optimal settings of $k = 5$, distance-weighted voting, ball-tree algorithm, and Minkowski distance. The trained model achieved an f1-score of 97.7% on the training dataset and 77.7% on the test dataset. The martensite-retained austenite ({M, RA}) class was identified as the most challenging to predict, with an f1-score of only 51%, attributable to its inherently complex and sensitive formation conditions.

Model insight analysis revealed that manganese content and Q&P temperatures are the most significant parameters governing microstructure evolution. The model was also used in a re-engineering exercise to identify steel compositions and heat treatment windows for targeted {M, RA} microstructure. The experimental results closely corroborated the model predictions, confirming the utility of the k-NN approach as a tool for microstructure-informed steel design.

Keywords: *Quenched and Partitioned Steel, Machine Learning, k-Nearest Neighbor, Microstructure Classification, SMOTE, AHSS, Steel Design*

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Chapter 1

Introduction and Literature Review

1.1 Background

The design and development of advanced structural materials, particularly steels, relies heavily on understanding the relationship between microstructure and mechanical properties. Microstructure classification — the identification of phases or microconstituents present in a micrograph — is therefore a fundamental activity in materials science and engineering. Traditionally, this process demands significant expert knowledge and is highly time-consuming, motivating the search for automated and intelligent classification tools.

In the era of Industry 4.0, the rapid development of Artificial Intelligence (AI), machine learning (ML), and image processing technologies offers promising avenues to automate and accelerate materials characterisation. Large volumes of steel microstructure data accumulated over decades of research represent a rich repository that can be analysed using computational tools to extract actionable insights. Data-driven approaches have already demonstrated success in classifying microstructures of carbon steel, dual-phase (DP) steel, high-strength low-alloy (HSLA) steel, and multi-phase steel.

Among the most exciting new steel families is Quenched and Partitioned (Q&P) steel, a third-generation Advanced High-Strength Steel (AHSS) that combines high tensile strength with good ductility. The microstructure of Q&P steel is particularly complex, typically comprising martensite, retained austenite, and other secondary phases depending on the alloying composition and heat treatment conditions. Despite the growing research output on Q&P steels, no systematic machine learning analysis of the Q&P microstructure database existed prior to the work reviewed in this project.

1.2 Q&P Steel and Advanced High-Strength Steel

Advanced High-Strength Steels (AHSS) represent the contemporary frontier of automotive and structural steel development. The core challenge in AHSS design is

balancing the trade-off between strength and formability — properties that are often mutually exclusive in conventional steel grades. The third generation of AHSS seeks to overcome this limitation by engineering steels with specific multiphase microstructures that exploit transformation-induced plasticity (TRIP) effects.

The Quenching and Partitioning (Q&P) process, pioneered by Speer et al., is a heat treatment route designed to retain significant fractions of austenite at room temperature. The process involves: (1) austenitisation at a temperature above A_{c3} or between A_{c1} and A_{c3} ; (2) quenching to a temperature between the martensite start (M_s) and martensite finish (M_f) temperatures to form a controlled fraction of martensite; and (3) a partitioning step at a higher temperature to facilitate carbon diffusion from martensite into the surrounding austenite, thereby stabilising it against transformation during final cooling.

The resulting microstructure consists predominantly of martensite and retained austenite (RA), along with secondary phases such as bainite, ferrite, and carbides depending on the specific processing conditions. The retained austenite is critical to the TRIP effect, which contributes to energy absorption during deformation and enhances ductility. Q&P steels achieve tensile strengths exceeding 1000 MPa while maintaining elongation values of 15–20%, making them highly attractive for automotive lightweight structural applications.

1.3 Machine Learning in Microstructure Classification

Machine learning methods have been progressively applied to microstructure classification and materials data analysis. A number of key prior studies inform the approach reviewed in this project:

DeCost and Holm (2015) demonstrated that visual feature representations of micrographs could be used to compute microstructural fingerprints, and that Support Vector Machine (SVM) classifiers could automatically classify microstructures by comparing histograms of visual features. Chowdhury et al. (2016) showed that feature extraction and dimensionality reduction techniques could represent micrographs as

feature vectors, and that k-NN, voting, SVM, and random forest classifiers could achieve effective classification.

Gola et al. (2018) applied SVM-based data mining to distinguish microstructures of dual-phase steels, while Azimi et al. (2018) proposed a deep-learning framework using a fully convolutional neural network (FCNN) for low-carbon steel microstructure classification, achieving an accuracy of 93.94%. Arivazhagan et al. (2019) applied k-NN and SVM for plain carbon steel microstructure classification using a novel texture descriptor called the Modified Alternate Local Ternary Pattern (MALTP). Kim et al. (2022) used the Bayesian-Gaussian mixture model with image filtering for unsupervised microstructure segmentation.

Despite this extensive work on other steel families, the machine learning-based analysis of Q&P steel microstructure had not been conducted before the work of Gupta et al. (2023), which forms the basis of this project.

1.4 Objectives of the Study

The primary objectives of this minor project are:

- To review and understand the machine learning-based classification framework applied to Q&P steel microstructure types.
- To study the k-NN algorithm and its implementation for multi-class classification problems in materials science.
- To analyse the dataset features, preprocessing steps including SMOTE, and the model evaluation methodology.
- To understand model insights into the effect of steel composition and heat treatment parameters on microstructure evolution.
- To review the re-engineering exercise and experimental validation carried out in the original work.
- To draw conclusions on the applicability and limitations of the k-NN model for Q&P steel microstructure prediction.

Chapter 2

Theoretical Study

2.1 Quenching and Partitioning (Q&P) Heat Treatment

The Quenching and Partitioning process is a thermomechanical heat treatment designed to produce a multiphase microstructure with significant retained austenite in steel. The process involves three main stages:

Stage 1: Austenitisation

The steel is heated to the full austenitisation temperature (above A_{c3}) or into the intercritical region (between A_{c1} and A_{c3}) and held for a sufficient time to form a homogeneous austenite (or austenite + ferrite) structure. The A_{c1} and A_{c3} temperatures depend on the steel's chemical composition and heating rate.

Stage 2: Quenching

The steel is rapidly cooled (quenched) to a temperature (Q_T) between the martensite start temperature (M_s) and the martensite finish temperature (M_f). The fraction of martensite formed is controlled by this quenching temperature, with lower Q_T producing more martensite. The remaining untransformed austenite is retained for the next stage.

Stage 3: Partitioning

The steel is held at (one-step Q&P: $Q_T = P_T$) or reheated to a partitioning temperature (P_T , usually above M_s). During this stage, carbon atoms diffuse from the supersaturated martensite into the surrounding austenite. This carbon enrichment lowers the M_s of the austenite, stabilising it against martensitic transformation during the final cooling to room temperature. At the end of partitioning, the steel is cooled to room temperature.

The critical transformation temperatures used in the dataset were estimated using established empirical formulas. The lower critical temperature A_{c1} and upper critical

temperature A_{c3} were calculated following Trzaska's formula, while the martensite start temperature M_s was estimated using the Andrews empirical formula.

2.2 Microstructure Types in Q&P Steels

Based on the review of approximately 107 research articles covering the last 20 years of Q&P steel research, six distinct microstructure types were identified and used as classification labels in the machine learning model. Table 2.1 presents these microstructure types along with their descriptions and numerical codes used for computational convenience.

Code	Microstructure Type	Description
0	{M, B, RA}	Martensite, Bainite, and Retained Austenite
1	{M, F, B, RA}	Martensite, Ferrite, Bainite, and Retained Austenite
2	{M, F, RA}	Martensite, Ferrite, and Retained Austenite
3	{M, F, RA, C}	Martensite, Ferrite, Retained Austenite with Carbide
4	{M, RA}	Martensite and Retained Austenite
5	{M, RA, C}	Martensite and Retained Austenite with Carbide

Table 2.1: Q&P Steel Microstructure Types and Their Descriptions

The {M, RA} type microstructure is the most desirable in Q&P steels from a performance standpoint, as it offers the cleanest combination of hard martensite and transformation-capable retained austenite without the presence of softer phases (ferrite) or brittle phases (carbides). However, it is also the most difficult to achieve and predict, as small variations in composition or processing temperature can easily introduce secondary constituents.

2.3 The k-Nearest Neighbor (k-NN) Algorithm

The k-Nearest Neighbor (k-NN) algorithm is a non-parametric, supervised machine learning algorithm used for classification and regression tasks. Unlike parametric algorithms, k-NN makes no prior assumptions about the distribution of the data. Its underlying principle is simple: data points that are close to each other in the feature space are likely to belong to the same class.

The k-NN algorithm classifies a new data point by examining the k most similar (nearest) data points from the training dataset and assigning the class that is most common among those k neighbours (majority voting). The working principle can be summarised in the following steps:

Step 1 — Selection of k: The value of k determines how many nearest neighbours to consider. A small k leads to an unstable (noisy) model, while a large k increases computational cost and may over-smooth decision boundaries. $k = 5$ is the most commonly used value and was found optimal in this study.

Step 2 — Distance Computation: For each new data point, distances to all training points are computed using a distance metric. The Euclidean (Minkowski $p = 2$) distance is most common. Efficient algorithms such as k-D Tree and Ball Tree are used to reduce computational complexity.

Step 3 — Identifying Nearest Neighbours: The k training points with the smallest distances to the query point are identified and their class labels are recorded.

Step 4 — Majority Voting: The class label that appears most frequently among the k nearest neighbours is assigned to the query point. In this study, distance-weighted voting was used (closer neighbours contribute more).

The k-NN algorithm is well-suited to multi-class classification problems with moderate dataset sizes. Its primary advantages include simplicity, no training phase (lazy learner), and natural support for multiple classes without modification. A key limitation is that it stores the entire training set in memory and requires computing distances at inference time, which can be expensive for large datasets.

2.4 Model Evaluation Metrics

Since the problem is a multi-class classification task with class imbalance, multiple evaluation metrics were used beyond simple accuracy. The key metrics are defined below:

Precision

Precision measures the fraction of predicted positives that are truly positive. For each class, it is defined as:

$$Precision = TP / (TP + FP)$$

Recall

Recall measures the fraction of actual positives that are correctly identified:

$$Recall = TP / (TP + FN)$$

F1-Score

The F1-score is the harmonic mean of precision and recall, providing a single balanced measure of classifier performance:

$$F1-Score = 2 \times (Precision \times Recall) / (Precision + Recall)$$

The weighted F1-score was used as the primary performance metric in this study, as it accounts for class imbalance by weighting each class's F1-score by its support (number of instances). The confusion matrix was used as an additional diagnostic tool to visualise the pattern of correct and incorrect predictions for each class.

Chapter 3

Dataset and Methodology

3.1 Dataset Compilation

The construction of a high-quality, representative dataset was identified as the most critical and challenging initial step in the machine learning exercise. To create this dataset, an extensive literature survey was conducted covering approximately 107 research articles published over the last 20 years on Q&P steel processing, microstructure, and properties. This survey yielded information on approximately 400 microstructures of Q&P steels.

Each data entry was subjected to manual investigation for inconsistencies and ambiguities. After cleaning, a final dataset of 348 samples (rows) and 16 compositional and processing parameters (columns) was obtained. Subsequent significance analysis led to the removal of certain parameters with insufficient variability or relevance, including sulphur (S wt.%), phosphorus (P wt.%), and heat treatment time parameters.

Microalloying elements — Niobium (Nb), Vanadium (V), and Titanium (Ti) — were found to be inconsistently reported across the literature, making it impractical to treat each as an independent feature. To preserve their collective metallurgical significance without losing information, a new derived feature was created: the Total Micro-Alloying Element content ($\text{TMAE} = \text{wt.\% Nb} + \text{wt.\% V} + \text{wt.\% Ti}$).

3.2 Feature Engineering

The final processed feature matrix had dimensions 348×9 . The nine input features (independent variables) are described in Table 3.1:

Feature	Description	Min	Max	Mean	SD
C (wt.%)	Carbon content	0.03	1.20	0.29	0.18
Si (wt.%)	Silicon content	0.00	2.60	1.35	0.54
Mn (wt.%)	Manganese content	0.24	6.00	2.11	1.18

TMAE (wt.%)	Total microalloying elements	0.00	0.30	0.02	0.05
Ac1 (°C)	Lower critical temperature	693.87	772.50	742.42	13.01
Ac3 (°C)	Upper critical temperature	703.28	916.44	829.26	35.76
Ms (°C)	Martensite start temperature	120.00	453.60	359.01	61.23
QT (°C)	Quenching temperature	10.00	550.00	228.50	76.29
PT (°C)	Partitioning temperature	150.00	650.00	367.09	80.29

Table 3.1: Statistical description of the 9-dimensional feature space

The Ac1, Ac3, and Ms temperatures were not directly measured in all cases; they were estimated from the chemical composition using established empirical formulae (Trzaska's formula for Ac1 and Ac3; Andrews' formula for Ms). This ensured completeness of the dataset and physical consistency with the alloy compositions.

3.3 Data Preprocessing

3.3.1 Feature Standardisation

The nine features span very different numerical ranges — from small decimal fractions for compositional features to hundreds of degrees Celsius for temperature features. Without scaling, distance-based algorithms such as k-NN would be dominated by the temperature features, leading to biased learning. Therefore, all features were standardised to zero mean and unit variance using the z-score normalisation:

$$x'_i = (x_i - \mu) / \sigma$$

where x'_i is the standardised feature value, x_i is the original value, μ is the mean, and σ is the standard deviation of the feature column.

3.3.2 Class Imbalance and SMOTE

Exploratory analysis of the class distribution revealed a significant imbalance: the {M, RA} and {M, B, RA} classes were dominant, while the {M, F, B, RA} and {M, F, RA, C} classes were severely underrepresented. Training a classifier on such imbalanced data leads to biased predictions favouring the majority classes.

To address this, the Synthetic Minority Over-sampling Technique (SMOTE) was applied. SMOTE generates synthetic data points for minority classes by interpolating

between existing minority class samples in the feature space. Specifically, for each minority class sample, its k-nearest neighbours (within the same class) are identified, and new synthetic samples are created along the line segments connecting the sample to its neighbours. SMOTE upsampled each class to 146 samples (the size of the largest class in the original dataset), resulting in a balanced training dataset.

3.4 Model Development

3.4.1 Train-Test Split

The preprocessed and SMOTE-balanced dataset was partitioned in a 70/30 ratio: 70% of the data was used for training the k-NN classifier, and the remaining 30% was held out as an independent test set for unbiased performance evaluation. The test set was not subjected to SMOTE, ensuring that evaluation reflects performance on real (not synthetic) data.

3.4.2 k-Value Selection and Hyperparameter Tuning

The performance of the k-NN classifier is highly sensitive to the choice of k and other hyperparameters. To identify the optimal k, the classifier was first trained and evaluated (using the test set f1-score) across a range of k values from 2 to 16. A k = 5 (the standard default value) was found to yield the best trade-off between training and test performance.

A comprehensive hyperparameter grid search was then conducted over the following parameter space using scikit-learn's GridSearchCV:

- n_neighbors: [3, 5, 7, 9, 11, 13, 15]
- weights: ['uniform', 'distance']
- algorithm: ['ball_tree', 'kd_tree', 'brute']
- leaf_size: [15, 20, 25, 30]
- p (Minkowski metric): [1, 2]

The optimal hyperparameter configuration was found to be: n_neighbors = 5, weights = 'distance', algorithm = 'ball_tree', leaf_size = 20, p = 2 (Euclidean distance). The use

of distance-weighted voting gives closer neighbours proportionally more influence on the predicted class than distant neighbours.

Chapter 4

Results and Discussion

4.1 Model Performance on Training and Test Data

The k-NN classifier with optimised hyperparameters ($k = 5$, distance-weighted, ball-tree algorithm) was evaluated on both the training and test datasets using the f1-score as the primary performance metric.

- Training dataset f1-score: 97.7%
- Test dataset f1-score: 77.7%

The substantial gap between training and test performance is typical of a k-NN classifier, which has no generalisation mechanism during training (it simply memorises the training data). The test performance of 77.7% is nonetheless considered strong for a six-class classification problem on a dataset of only 348 samples compiled from diverse sources.

The evolution of f1-score with k-value on the test dataset showed that $k = 5$ yielded the peak performance. Values of k below 5 resulted in overfitting (high train, lower test scores), while higher k values led to progressively poorer test performance as the classifier could no longer differentiate closely-spaced class boundaries.

4.2 Classification Report and Confusion Matrix

The per-class performance metrics on the test dataset are presented in Table 4.1:

Microstructure Type	Precision	Recall	F1-Score	Support
{M, B, RA}	0.80	0.77	0.79	48
{M, F, B, RA}	0.83	0.89	0.86	45
{M, F, RA}	0.79	0.75	0.77	36
{M, F, RA, C}	0.87	0.91	0.89	45
{M, RA}	0.57	0.47	0.51	43
{M, RA, C}	0.77	0.89	0.83	46

Macro Average	0.77	0.78	0.78	263
Weighted Average	0.78	0.78	0.78	263

Table 4.1: Classification report on the test dataset

Five of the six microstructure classes achieved f1-scores above 0.75, indicating strong predictive capability. The {M, F, RA, C} class achieved the highest f1-score of 0.89, followed by {M, F, B, RA} at 0.86 and {M, RA, C} at 0.83.

The {M, RA} class is a notable exception, achieving a precision of 0.57, recall of 0.47, and f1-score of only 0.51. Confusion matrix analysis revealed that out of 43 actual {M, RA} cases in the test data, the model correctly predicted only 20. The remaining 23 cases were misclassified as: {M, B, RA} (8 cases), {M, F, B, RA} (5 cases), {M, RA, C} (4 cases), {M, F, RA} (3 cases), and {M, F, RA, C} (3 cases).

This high confusion rate is metallurgically explainable: achieving a pure {M, RA} microstructure requires precise control of both chemistry and processing to exclude secondary phases. In practice, trace amounts of bainite, ferrite, or carbides are common. The subtle compositional and process differences that distinguish {M, RA} from adjacent microstructure types create an inherently ambiguous classification boundary.

4.3 Model Insight: Effect of Parameters on Microstructure

To extract physically meaningful insights, a systematic sensitivity analysis was performed. For each feature, a synthetic dataset was generated by varying that feature across its full range while holding all other features at their mean values. The k-NN model was then used to predict the microstructure type for each synthetic data point.

Effect of Carbon Content (C wt.%)

Low carbon content favours the formation of {M, F, RA} type microstructure, reflecting the tendency of low-carbon steels to retain ferrite regions during intercritical austenitisation or Q&P processing. As carbon content increases, the microstructure transitions toward the {M, RA} type, consistent with the increased hardenability and greater martensite stability in higher-carbon compositions.

Effect of Silicon Content (Si wt.%)

Silicon below 0.5 wt.% tends to produce {M, B, RA} microstructure, as insufficient silicon fails to suppress cementite precipitation during the partitioning step, leading to bainite formation. In the range of 0.75 to 2.25 wt.% Si, a high probability of {M, RA} type microstructure formation is predicted, consistent with silicon's well-known role as a carbide-suppressing element in Q&P steels.

Effect of Manganese Content (Mn wt.%)

Manganese was identified as the most influential compositional feature. The model predicts {M, RA} type microstructure for Mn up to 4 wt.%, while Mn in the range of 4 to 6 wt.% is associated with {M, F, RA} microstructure. This behaviour is related to the Mn partitioning effect: at high Mn levels, the redistribution of Mn between phases during heat treatment changes the relative stability of austenite and ferrite, promoting ferrite formation.

Effect of Quenching Temperature (QT)

Low quenching temperatures (below 300°C) tend to produce {M, RA} type microstructure, as quenching to a low temperature maximises martensite formation. Quenching temperatures above 300°C can produce {M, B, RA} type, as the less complete quench leaves more untransformed austenite that partially transforms to bainite during subsequent partitioning.

Effect of Partitioning Temperature (PT)

Partitioning temperatures below 200°C are predicted to produce carbide-free {M, RA} microstructures, as insufficient thermal energy prevents carbide precipitation. Partitioning in the 200–600°C range is associated with the presence of carbides in the microstructure ({M, RA, C} or {M, F, RA, C}). At very high partitioning temperatures, a tendency toward {M, B, RA} formation is observed.

4.4 Re-engineering for Target Microstructure

A computer experiment was designed to identify the composition and heat treatment parameter window that would reliably produce the target {M, RA} microstructure. A full-factorial synthetic dataset was generated with:

- C: five levels — 0.03, 0.32, 0.615, 0.91, 1.20 wt.%
- Si: five levels — 0.04, 0.68, 1.32, 1.96, 2.60 wt.%
- Mn: five levels — 1.00, 1.75, 2.50, 3.25, 4.00 wt.%
- TMAE: three levels — 0, 0.016, 0.036 wt.%

k-NN model predictions on this $5 \times 5 \times 5 \times 3$ dataset identified a consistent {M, RA} prediction region at approximately C = 0.61 wt.%, Si in the range 0.68–1.96 wt.%, and Mn in the range 1.0–4.0 wt.%. A target composition was proposed as C = 0.70 wt.%, Si = 1.8 wt.%, Mn = 2.0 wt.%, with TMAE = 0.036 wt.%.

Heat treatment parameter mapping for this composition showed a potential {M, RA} processing window with Qtemp and Ptemp in the range of 150–230°C, particularly for the higher TMAE level of 0.036 wt.%, where a broader stable zone was identified.

Chapter 5

Conclusions and Future Scope

5.1 Conclusions

This project presented a detailed review and study of a k-nearest neighbor (k-NN) machine learning model for the multi-class classification of Quenched and Partitioned (Q&P) steel microstructure types. The following key conclusions are drawn:

1. A comprehensive Q&P steel microstructure database was compiled from 107 research articles, yielding 348 samples characterised by nine features covering chemical composition and heat treatment parameters. Six distinct microstructure types were identified and labelled.
2. SMOTE oversampling effectively addressed the class imbalance problem inherent in the collected data, improving the classifier's ability to learn minority-class boundaries.
3. The k-NN classifier achieved an overall f1-score of 97.7% on the training dataset and 77.7% on the independent test dataset. Five of the six microstructure classes were predicted with f1-scores above 0.75, demonstrating strong generalisation.
4. The {M, RA} microstructure class was identified as the most difficult to predict (f1-score = 0.51), reflecting its inherently high sensitivity to small variations in composition and processing. Any model prediction of {M, RA} should be regarded with caution and verified experimentally.
5. Model sensitivity analysis identified Mn (wt.%) content and the quenching and partitioning temperatures as the most dominant parameters governing microstructure evolution in Q&P steels. Silicon content was found to be critical in suppressing carbide precipitation and enabling {M, RA} formation.
6. The re-engineering exercise using the k-NN model successfully identified a compositional window ($C \approx 0.70$ wt.%, $Si \approx 1.8$ wt.%, $Mn \approx 2.0$ wt.%, $TMAE \approx 0.036$ wt.%) and processing window ($QT/PT \approx 150\text{--}230^\circ\text{C}$) for targeted {M, RA} microstructure production.

5.2 Future Scope

The work reviewed in this project opens several directions for future research and development:

- Expanding the dataset: Inclusion of more data points, especially for underrepresented microstructure classes, would improve model robustness and reduce prediction uncertainty for difficult classes like {M, RA}.
- Advanced ML algorithms: Comparison with ensemble methods (Random Forest, Gradient Boosting, XGBoost) and deep learning approaches (Artificial Neural Networks) may yield improved classification accuracy.
- Feature importance analysis: Techniques such as SHAP (SHapley Additive exPlanations) values could provide more rigorous, quantitative insights into the relative importance of each input feature.
- Image-based classification: Incorporating optical and SEM micrograph image data (alongside tabular composition and processing data) as additional inputs to a hybrid ML model could significantly improve prediction of complex microstructures.
- Extension to other steel grades: The modelling framework could be extended to other AHSS families (e.g., medium Mn steel, nanostructured bainitic steel) for broader microstructure prediction capability.
- Online process control: Integration of the trained model into an online heat treatment monitoring and control system for real-time microstructure prediction during industrial Q&P processing.
- Uncertainty quantification: Implementing probabilistic classifiers (e.g., Gaussian Process Classifiers or Bayesian k-NN) to provide confidence intervals alongside microstructure predictions, particularly important for the uncertain {M, RA} class.

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